J Stomatol Oral Maxillofac Surg 125 (2024) 101948

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Original Article

Comparison of temporomandibular joint osseous components in unilateral and bilateral cleft lip and palate patients and normal controls: A CBCT study

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A R T I C L E I N F O

Article History:
Received 15 April 2024
Accepted 15 June 2024
Available online 17 June 2024

Keywords:
Cleft lip and palate
Temporomandibular joint
Cone beam computed tomography

A B S T R A C T

Objective: The objective of this study was to conduct a comparative analysis of the components of the temporomandibular joint in individuals with unilateral, bilateral cleft lip and palate (CLP), and in healthy individuals, utilizing cone beam computed tomography (CBCT) images.
Method and material: The present study employed a cross-sectional design and recruited participants aged 18 to 30 years. The participants were categorized into three groups: a control group consisting of 36 individuals without any cleft, a group of 35 patients with unilateral cleft lip and palate (UCLP), and a group of 15 patients with bilateral cleft lip and palate (BCLP). The analysis of CBCT images encompassed the examination of condylar height and angulation, glenoid fossa height and width, articular eminence inclination, as well as joint spaces across all three groups. The Kruskal-Wallis and Mann-Whitney tests were employed to ascertain the significant differences among the three groups.
Results: The UCLP and BCLP groups demonstrated a statistically significant reduction in condylar height and articular eminence inclination in comparison to the control group. Furthermore, a significant difference in the width of the glenoid fossa was seen between the group with clefts and the control group.
Conclusion: The CBCT images showed significant differences in several aspects of the temporomandibular joint, including condylar height, articular eminence inclination, and glenoid fossa width, in individuals with cleft palate. These abnormalities can contribute to the development of temporomandibular joint diseases. Therefore, recognizing these distinctions can help prevent further deterioration and progression of temporomandibular disorders (TMD) in CLP patients.

1. Introduction

The human palate is comprised of two separate components: the hard palate, which is constructed of bone, and the soft palate, situated posteriorly to the hard palate [1]. The formation of the lips, hard palate, and soft palate occurs as a result of the fusion of craniofacial features in the process of embryological development. The potential for the formation of cleft lip and palate (CLP) exists when there is any interruption in these processes. [2]. CLP, a prevalent congenital deformity in the orofacial area, occurs at a prevalence of 0.45 per 1000 live births [3]. Clefts can be categorized as either isolated or nonsyndromic, or syndromic, depending on the presence or absence of other serious deformities [4,5].

Individuals diagnosed CLP exhibit a greater occurrence of dental malformations, including dental agenesis, microdontia, peg-shaped front teeth, supernumerary teeth, impacted teeth, and tooth malposition, etc. [6,7]. Furthermore, there is evidence to suggest that the developmental trajectory of dentomaxillofacial tissues in individuals with cleft lip and palate (CLP) diverges from that of individuals without this congenital anomaly [8]. In patients with cleft lip and palate (CLP), the mandible is the second most afflicted component of the craniofacial apparatus, after the maxilla [9]. Cranial base structures, like the temporomandibular fossa, and the condylar development are connected to the location and development of the mandible. Modifications to these structures have the potential to induce changes in the growth pattern of the mandible, resulting in mandibular asymmetry and eventual malocclusion [10]. Furthermore, it should be noted that the facial muscles in individuals with CLP have an asymmetrical function, which has the potential to exacerbate any previous malocclusions, particularly posterior unilateral crossbite [11-13]. It is

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https://doi.org/10.1016/j.jormas.2024.101948

reasonable to anticipate that modifications to the components of the temporomandibular joint (TMJ) may occur as a result of the various dental and occlusal changes [13,14]. The TMJ undergoes ongoing remodeling as an adaptive reaction. However, if the joint's capacity to withstand high loading forces through remodeling is exceeded, degenerative alterations occur, potentially resulting in temporomandibular disorder (TMD) [15-17].

TMD refers to a collection of orofacial disorders that are distinguished by discomfort experienced in the preauricular region, temporomandibular joint (TMJ), or the muscles responsible for mastication. Additionally, TMD is associated with limits and deviations in the mandible's range of motion, as well as auditory manifestations in the TMJ during jaw movement [13,18]. Individuals diagnosed CLP exhibit a greater prevalence of temporomandibular joint (TMJ) dysfunction compared to individuals without CLP. For instance, a study has documented that individuals with CLP demonstrated a notable decrease in their jaw-opening movement in comparison to a group of persons who did not have CLP [19]. The clinical manifestations of temporomandibular disorder (TMD) are impacted by various risk factors [15]. Numerous ideas have been postulated in order to elucidate the etiology of temporomandibular disorders (TMD). These theories encompass untreated malocclusions, unstable occlusions, traumatic events, individual predispositions, and anatomical abnormalities [18]. The presence of distinct dentomaxillofacial growth patterns and malocclusion in patients with CLP may potentially contribute to the alteration of the articular structures of the TMJ and the resorption of the condylar articular surface. These factors collectively increase the likelihood of temporomandibular joint disorder development in these patients [10,13]. Furthermore, TMD exhibits a significant psychosocial element, often accompanied by various comorbid psychiatric conditions such as depression and anxiety. Individuals who are born with craniofacial defects, such as CLP, which result in facial disfigurement, are believed to have heightened susceptibility to psychiatric disorders and increased vulnerability to stress in their later years, hence rendering them more predisposed to TMD [9,20].

To the best of what we know at present, there exists an insufficient amount of research that has examined several skeletal components of the TMJ in patients with CLP [10,13,21,22]. Given the ongoing debate around the outcomes of previous investigations, The aim of this study was to evaluate and juxtapose several elements of the TMJ in individuals with UCLP and BCLP conditions, as well as in those without such conditions, using CBCT images.

2. Materials and method

The current investigation received approval from the Institutional Research Committee under the reference number IR.SUM.DENTAL.REC.1399.09-08 with the clinical trial number of 2991. In this study, researchers collected patients' data and CBCT images from an oral and maxillofacial radiology clinic. CBCT images were taken for various purposes like implant placement and orthodontic interventions. a control group of 36 individuals without CLP (17 females and 19 males; mean age 22.27± 4.883), a group of 35 patients with unilateral cleft lip palate (UCLP) (16 females, 19 males; mean age 22.09± 4.58), and a group of 15 patients with bilateral cleft lip palate (BCLP) (6 females and 9 males; mean age 22.00 ± 4.76).

Exclusion criteria included patients who had clinical manifestations of TMD, exhibited facial asymmetry during clinical examination, had a documented history of trauma, were undergoing orthodontic or prosthodontic treatment, presented with craniofacial abnormalities, or had any underlying systemic problems. Ages and sexes were matched in both cleft and control group.

The CBCT images were obtained at the same radiography center and supervised by the same clinician. Using a New Tom VGi (NewTom, Verona, Italy), CBCT images of the both right and left temporomandibular joints were acquired. The imaging parameters used were 110 (KVp), 3.05 (mA), and 3.6 s of exposure time. The images were captured at a standard resolution (voxel size of 0.3). The image field dimensions were 15 by 15 cm. Patients assumed an upright position and displayed maximal intercuspal occlusion while clenching their teeth. It was assured that the Frankfort plane was parallel to the floor while the subjects' heads were positioned. For the preparation of the TMJ images, a NewTom Cone Beam 3D imaging system workstation (NNT Software version 6.2) was utilized. The evaluation of the CBCT scan was performed in a specialized reporting room that offered optimal viewing conditions, specifically a dimly illuminated atmosphere. The scan was reconstructed and displayed on a Barco-China high-resolution monitor.

During the reconstruction procedure, the original data were recreated exclusively for the TMJ. For secondary reconstruction, the axial image with the largest mediolateral diameter of the condylar process was selected by scrolling axial images. In the selected axial view, a line parallel to the longitudinal axis of the condylar process was traced. Subsequently, lateral slices with a slice interval of 0.5 mm and a slice thickness of 0.5 mm were reconstructed.

On central sagittal section the fallowing parameters were measured:

The mandibular condyle height: was defined as the part of the mandible above the condylar plane (line parallel to the FH plane passing through the sigmoid notch (Fig. 1).

The height of the glenoid fossa: was established by measuring the perpendicular distance between the highest point of the fossa and the line that intersects the most inferior point on the articular eminence and the posterior glenoid (Fig.2).

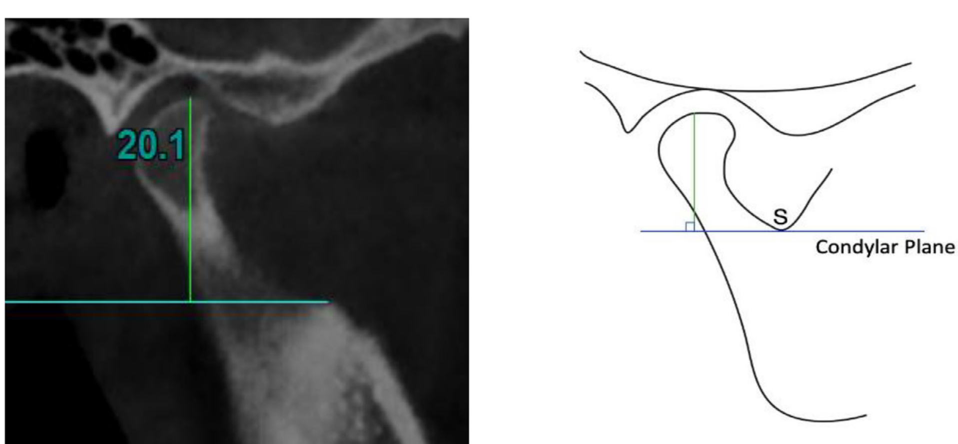

Fig. 1. Measurement of the condylar height; a part of the mandible above the condylar plane (line parallel to the FH plane passing through the sigmoid notch (S)).

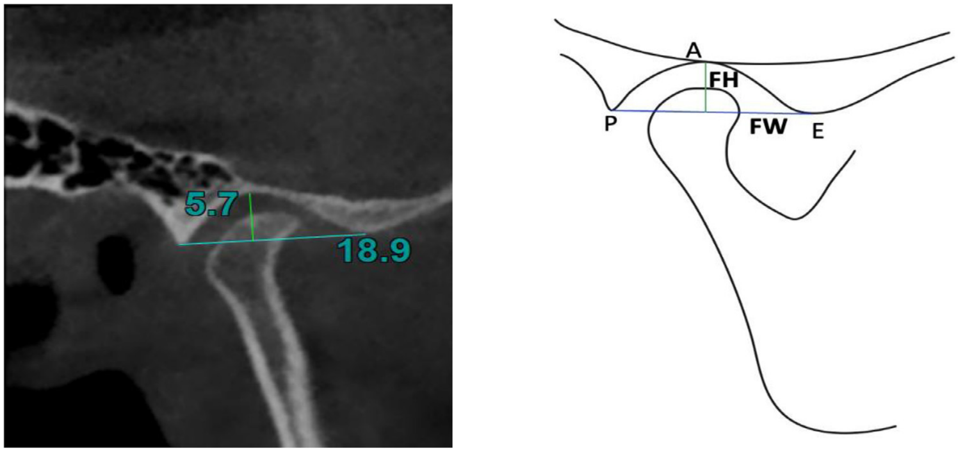

Fig. 2. Measurement of the height of the glenoid fossa (FH): the perpendicular distance between the highest point of the fossa (A) and the line that intersects the most inferior point on the articular eminence (E) and the posterior glenoid process (P).Measurement of the width of the glenoid fossa (FW): the distance between the most inferior point on the articular eminence (E) and the posterior glenoid process (P).

The width of the glenoid fossa: was defined as the distance between the most inferior point on the articular eminence and the posterior glenoid process (Fig.2).

The articular eminence inclination: The angle between the Frankfort plane and the plane passing through the highest point in the roof of glenoid fossa and the lowest point at the crest of the articular eminence (Fig.3).

Anterior, posterior and superior joint spaces: the values of the narrowest posterior (P) and anterior (A) superior joint (S) space were measured (Fig.4).

The condylar Angulation: the angle between the condylar axis (i.e. a line perpendicular to the long axis of the condylar Process) and the sagittal plane. Note that this parameter was measured on the axial view, in which the condylar process had the widest mediolateral diameter (Fig.5).

All CBCT images were measured by a researcher who was blind to the patient groups. To assess the presence of significant observer errors during measurement, the same operator remeasured all CBCT images two weeks later. The intraclass correlation coefficient (ICC) was utilized to evaluate the congruence between the initial and subsequent assessments. To evaluate the interoperator method error, a random sample comprising roughly 30 % of the acquired data was chosen and afterwards re-measured by an additional observer. The final data were subsequently subjected to comparison through the utilization of intraclass correlation coefficients.

The statistical analysis was performed utilizing SPSS version 22 (SPSS Inc., Chicago, IL) with a significance level of P 0.05. The Kruskal-Wallis and Mann-Whitney tests were used to determine if there were statistically significant differences between the three categories.

3. Results

The resulting interclass correlation coefficient (ICC) value of 0.967 indicates a substantial level of agreement between the initial and subsequent measurements, thereby validating the reliability of the measures. Furthermore, a notable degree of agreement was noted among the operators for all measurements. (ICC $\geq$0.90).

The comparison between the CLP group (BCLP $\pm$ UCLP) and the control group revealed a statistically significant difference in condylar height, articular eminence inclination, and glenoid fossa width. The control group exhibited significantly higher values for these three variables (Table 1).

Table 2 presents the descriptive statistics and comparative analysis of the variables across three distinct groups. The control group exhibited considerably greater condylar height and articular eminence inclination compared to both the UCLP and BCLP groups.

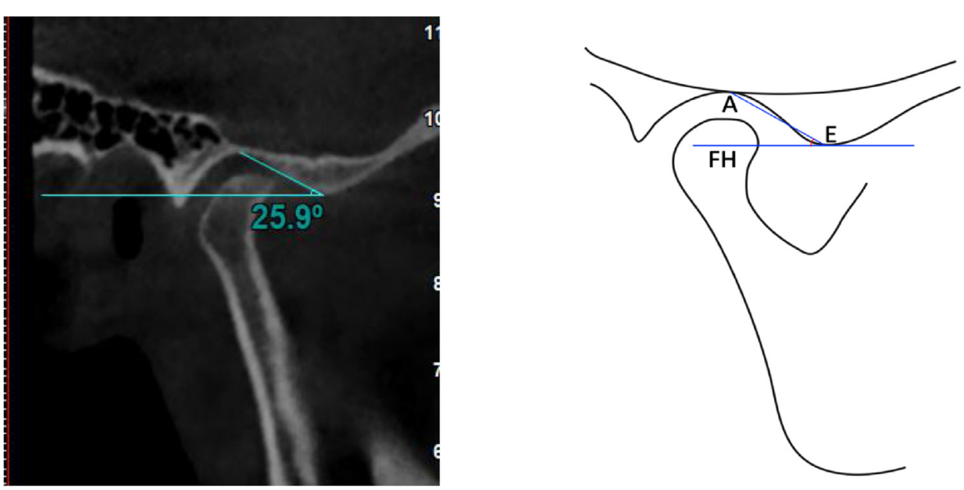

Fig. 3. Measurement of the articular eminence inclination: The angle between the Frankfort plane (FH) and the plane passing through the highest point in the roof of glenoid fossa (A) and the lowest point at the crest of the articular eminence (E).

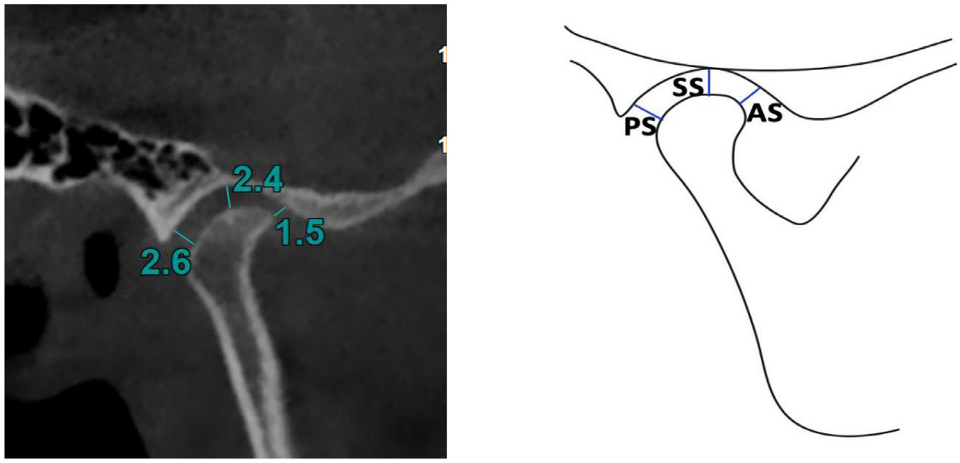

Fig. 4. Measurement of the subjective closest joint spaces; anterior joint space (AS), superior joint space (SS) and posterior joint space (PS).

Nevertheless, a notable disparity was not observed between the groups with UCLP and BCLP.

4. Discussion

Cleft lip and palate is widely recognized as the most frequently encountered congenital craniofacial deformity. According to research findings, individuals with cleft lip and palate have a higher prevalence of dental malformations, periodontal complications, occlusal interferences, and temporomandibular joint dysfunction [23,24]. The objective of this study was to assess the effects of CLP on the different elements of the temporomandibular joint, employing CBCT images as the primary source for analysis. The results indicate that cleft lip and palate have a discernible influence on specific aspects of the temporomandibular joint (TMJ), such as condylar height, inclination of articular eminence, and glenoid fossa width.

This study observed that patients diagnosed with UCLP and BCLP exhibited a reduced condylar height in comparison to the control group. Nevertheless, several prior studies have reported that there is no statistically significant dissimilarity in condylar height among individuals diagnosed with CLP [10,21,22]. The discrepancy in findings can potentially be attributed to factors such as variations in sample size, the age range of participants, and divergent measurement methodologies employed in these investigations.

Fig. 5. the angle between the condylar axis (i.e. a line perpendicular to the long axis of the condylar Process) and the sagittal plane.

One potential rationale for the observed reduction in condylar height among the people with cleft in this study may stem from a modified maxillofacial growth pattern relative to the non-cleft individuals. The existence of scar tissue resulting from surgical procedures for cleft lip and/or palate notably impacts craniofacial development. Surgical complications such as reduced alveolar arch size, midface retrusion, and mandibular hypoplasia together with dental anomalies as a prevalent characteristic among cleft patients can potentially lead to the development of malocclusion [25,26].

The sensitivity of condyles to occlusal alterations is well-documented in academic literature. Hence, in persons afflicted with orofacial clefts, the presence of maxillary constriction and dental abnormalities can lead to malocclusion, resulting in improper stresses exerted on the condyles. Therefore, it may be inferred that the condyles and the temporomandibular joint will experience adaptive changes, resulting in a modified pattern of mandibular growth [27,28].

The findings of this study indicate a notable decrease in the width of the glenoid fossa in individuals diagnosed with cleft lip and palate (UCLP and BCLP), in comparison to the control group. The observed variation in size may be attributed to the relatively smaller condyles found in people with clefts, a finding that has been also reported by F. Amiri et al. and Talaeipour AR, et al. [13,21]. Furthermore, F.I. Ucar et al. discovered a notable decrease in condylar volume among the cleft group [10]. This finding provides additional support for the outcomes of the present investigation.

This study revealed that patients with UCLP or BCLP had lower articular eminence inclination than the control group which is in contrast to the results of F.I. Ucar et al. [1].

Table 1
comparison of the TMJ components between cleft and control groups.

Variables	Groups		
	Control (mean ± SD)	CLP* (mean ± SD)	P value
Condylar angle (°)	22.243 ± 7.7761	22.568 ± 10.7785	0.779
Condylar height (mm)	17.26 ± 3.391	15.52 ± 3.264	0.007**
Glenoid fossa height (mm)	6.379 ± 1.3655	6.829 ± 8.2812	0.062
Glenoid fossa width (mm)	17.78 ± 1.844	16.93 ± 2.292	0.012**
Articular eminence inclination (°)	37.682 ± 9.4335	30.137 ± 10.0042	0.000**
Anterior joint space (mm)	2.043 ± 0.6807	1.996 ± 0.6446	0.906
Superior joint space (mm)	3.456 ± 1.0173	3.284 ± 0.8035	0.812
Posterior joint space (mm)	2.401 ± 0.8196	2.381 ± 0.8042	0.923

* cleft lip palate patients.
** significant result.

Table 2
Comparison of the TMJ variables among UCLP, BCLP and control groups.

variables	Groups			P value
	Control (mean ± SD)	UCLP* (mean ± SD)	BCLP** (mean ± SD)	
Condylar angle (°)	21.850 ± 7.3800	22.680 ± 12.8598	21.193 ± 8.0036	0.953
Condylar height (mm)	17.525 ± 3.2346	15.320 ± 2.7765	15.820 ± 2.5744	0.027***
Glenoid fossa height (mm)	6.392 ± 1.3668	5.989 ± 1.6752	6.387 ± 1.9989	0.379
Glenoid fossa width (mm)	18.142 ± 1.8464	17.063 ± 2.2751	17.580 ± 2.8001	0.151
Articular eminence inclination (°)	36.906 ± 9.1983	29.506 ± 10.2253	28.100 ± 11.2283	0.005***
Anterior joint space (mm)	2.025 ± 0.6648	1.940 ± 0.5832	2.060 ± 0.6706	0.911
Superior joint space (mm)	3.350 ± 1.0734	3.131 ± 0.8109	3.220 ± 0.5545	0.751
Posterior joint space (mm)	2.403 ± 0.8895	2.340 ± 0.7361	2.253 ± 0.6947	0.985

* : Unilateral cleft lip palate patients.
** : Bilateral cleft lip palate patients.
*** : significant results.

The present investigation did not identify any statistically significant variations in terms of the condylar angle, glenoid fossa height, and joint spaces. Similarly, in a study conducted by Golshah et al. (29), it was observed that there were no statistically significant differences in the condylar axial angle, anterior joint space, superior joint space, and posterior joint space when comparing the UCLP groups with the control groups [29].

A study by Kim et al. found no significant differences in depth of the fossa and condylar angulation between both sides in patients with UCLP [30].

The research conducted by Ucar et al. and Talaeipour et al. yielded similar outcomes in terms of the height of the glenoid fossa in both the cleft group and the control group, which is consistent with the findings of our present investigation. However, they found lower condylar angulation for the cleft group [10,13].

Previous studies have employed various radiographic techniques, such as plain film radiography [31], conventional tomography [32], computed tomography (CT) [33,34], cone beam computed tomography (CBCT) [35,36], and magnetic resonance imaging (MRI) [37-39], to evaluate the morphology of the condyle. In the realm of conventional radiography, particularly in the case of panoramic radiography, it is important to acknowledge that the portrayal of the most lateral area of the condyle is constrained to a two-dimensional representation. Therefore, the use of these radiographs for assessing condylar morphology is less reliable. The limited accuracy in assessing alterations in condylar morphology can be related to the normal thickness of the slices used, which often falls within the range of 1 to 3 mm. MRI is a diagnostic modality utilized to assess the soft tissue constituents of the TMJ. The utilization of CT is widely prevalent within the medical domain as a diagnostic technique.

Nevertheless, the utilization of this technology in the field of dentistry is restricted due to various considerations, including limited availability, substantial costs, and potential hazards related to exposure to radiation. When doing a comparison between spiral CT and CBCT, it becomes evident that CBCT exhibits a spatial resolution that is either equivalent to or beyond that of spiral CT, achieving a degree of precision at or below the sub-millimeter range [39]. Therefore, in this study, CBCT was chosen as the most suitable imaging modality.

This investigation did not include a clinical examination of TMD signs and symptoms due to its retrospective nature. Further research with a larger cohort and concurrent clinical evaluations is needed to validate or challenge these findings.

5. Conclusion

The results of this study suggest that individuals CLP have alterations in a few features of temporomandibular joints, such as condylar height, articular eminence inclination, and glenoid fossa width. These modifications have the potential to contribute to the occurrence of TMD among individuals in this particular demographic. Therefore, by acknowledging these differences, it becomes feasible to prevent the worsening and progression of TMD in these individuals through timely identification and care.

Ethical approval

This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Ethics Committee of University Shiraz Dental School under the reference number IR.SUM.DENTAL.REC.1399.09–08. Written consent was taken from each patient for participation.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Availability of data and materials

Data and maretiala are available is requested.

Declaration of competing interest

The authors declared no conflict of interest.

CRediT authorship contribution statement

Salma Tabatabaei: Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Maryam Paknahad:** Writing – review & editing, Writing – original draft, Supervision. **Yalda Alamdarzadeh:** Writing – original draft, Supervision, Resources.

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